

Phase-Locking Value Analysis of High-Frequency Oscillations in Human Intracranial EEG

A Python Pipeline and Preliminary Findings

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Preprint Draft - April 2026

PRELIMINARY - NOT PEER REVIEWED

Abstract

ABSTRACT

High-frequency oscillations (HFOs) recorded from intracranial electroencephalography (iEEG) have emerged as promising biomarkers for epileptogenic tissue. While HFO rates are commonly studied, the functional connectivity patterns during HFO events remain less explored. Here we present an open-source Python pipeline for computing phase-locking values (PLV) between electrode pairs during validated HFO events in stereo-electroencephalography (SEEG) recordings.

We applied this pipeline to data from three patients undergoing presurgical evaluation for drug-resistant epilepsy. Our preprocessing steps included common average re-referencing restricted to good-quality SEEG channels, bandpass filtering in the HFO range (80-500 Hz), and Hilbert transform-based instantaneous phase extraction.

Across 13,912 validated HFO events from 27 recording sessions, we observed mean PLV values ranging from 0.577 to 0.761 across patients, with consistent connectivity patterns within individuals across recording runs. These preliminary findings suggest that phase synchronization during HFOs may carry patient-specific signatures.

Keywords: high-frequency oscillations, phase-locking value, SEEG, epilepsy, functional connectivity, Python

1. INTRODUCTION

High-frequency oscillations (HFOs) in the 80-500 Hz range have gained considerable attention as potential biomarkers of epileptogenic brain tissue (Jacobs et al., 2012; Zijlmans et al., 2017). These brief events, typically lasting 20-200 milliseconds, can be divided into ripples (80-250 Hz) and fast ripples (250-500 Hz), with the latter showing stronger association with seizure onset zones.

Most HFO research has focused on detection rates and spatial localization. However, growing evidence suggests that the functional relationships between brain regions during HFO events may provide additional clinically relevant information. Phase synchronization, quantified through metrics such as the phase-locking value (PLV), offers a way to assess transient functional connectivity that is independent of signal amplitude (Lachaux et al., 1999).

Recent work by Arnulfo et al. (2020) demonstrated that high-frequency oscillations can exhibit long-range phase synchronization across distributed cortical regions, challenging earlier assumptions about their purely local nature. This opens the question of whether phase synchronization patterns during pathological HFOs in epilepsy patients might differ from physiological activity.

Here we describe an open-source Python pipeline for HFO-based PLV analysis and report preliminary findings from four patients with drug-resistant epilepsy who underwent SEEG monitoring.

2. METHODS

2.1 Patients and recordings

We analyzed SEEG data from three patients undergoing presurgical evaluation for drug-resistant focal epilepsy at the University of Michigan. Electrode placement was determined solely by clinical criteria. All recordings were sampled at 4096 Hz.

Patient characteristics:

- sub-umich0020: 5 runs, 26 channels, 3,846 HFOs
- sub-umich0021: 11 runs, 46 channels, 4,590 HFOs
- sub-umich0022: 11 runs, 44 channels, 5,476 HFOs

2.2 HFO detection and validation

HFO events were detected using the qHFO algorithm (version 4.0, Staba detector). We retained only events occurring during periods when all good-quality SEEG channels simultaneously showed clean signal. Up to 500 validated HFOs per run were randomly selected for analysis.

2.3 Preprocessing pipeline

Step 1 - Common Average Re-referencing (CAR): Raw signals were re-referenced using the common average of all good-quality SEEG channels.

Step 2 - Bandpass filtering: Data were filtered to 80-500 Hz using a 10th-order elliptic filter with zero-phase filtering.

Step 3 - Phase extraction: Instantaneous phase was computed using the Hilbert transform applied to each channel independently.

Step 4 - PLV computation: For each HFO segment, PLV between channels i and j was computed as: $PLV = |\text{mean}(\exp(i * (\phi_i - \phi_j)))|$

PLV ranges from 0 (no phase coupling) to 1 (perfect phase locking).

3. RESULTS

3.1 HFO characteristics

Across the three patients, we analyzed 13,912 validated HFO events from 27 recording runs. Mean HFO duration was 35.6 ms (range: 6.8-516.8 ms). The number of good SEEG channels varied from 26 to 46 across patients.

3.2 Phase-locking values during HFOs

Mean PLV values varied substantially across patients:

- sub-umich0020: 0.761 +/- 0.027
- sub-umich0021: 0.665 +/- 0.039
- sub-umich0022: 0.577 +/- 0.045

3.3 Connectivity patterns

Visual inspection suggests patient-specific connectivity structure, with some electrode pairs showing consistently elevated PLV while others remain near baseline levels (see Figure 1).

3.4 Run-to-run consistency

Within each patient, mean PLV showed reasonable stability across recording runs. The coefficient of variation for run-level mean PLV ranged from 5% to 8% across patients, suggesting that the overall level of phase synchronization during HFOs is a relatively stable individual characteristic (see Figure 2).

Figure 1: Global Average PLV Connectivity Matrices

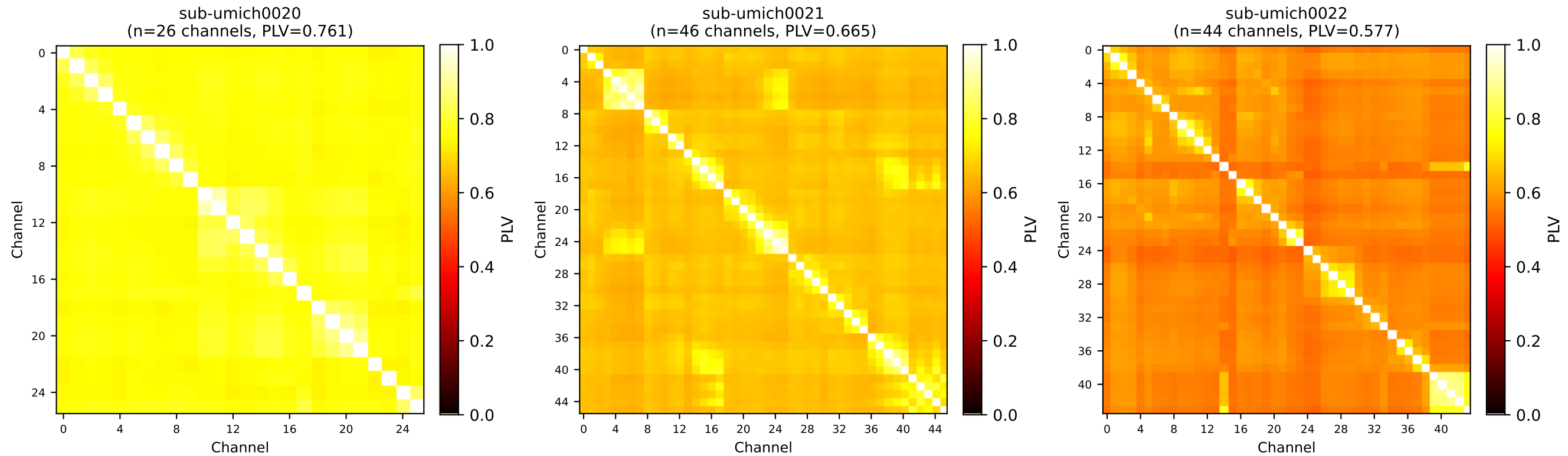


Figure 2: Mean PLV Across Recording Runs

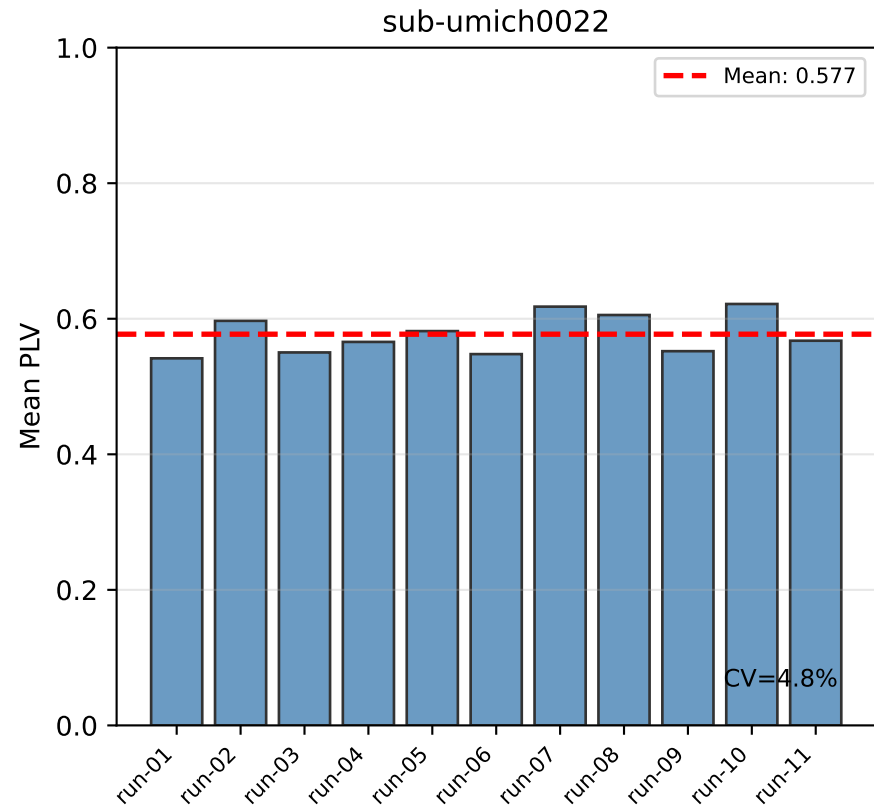
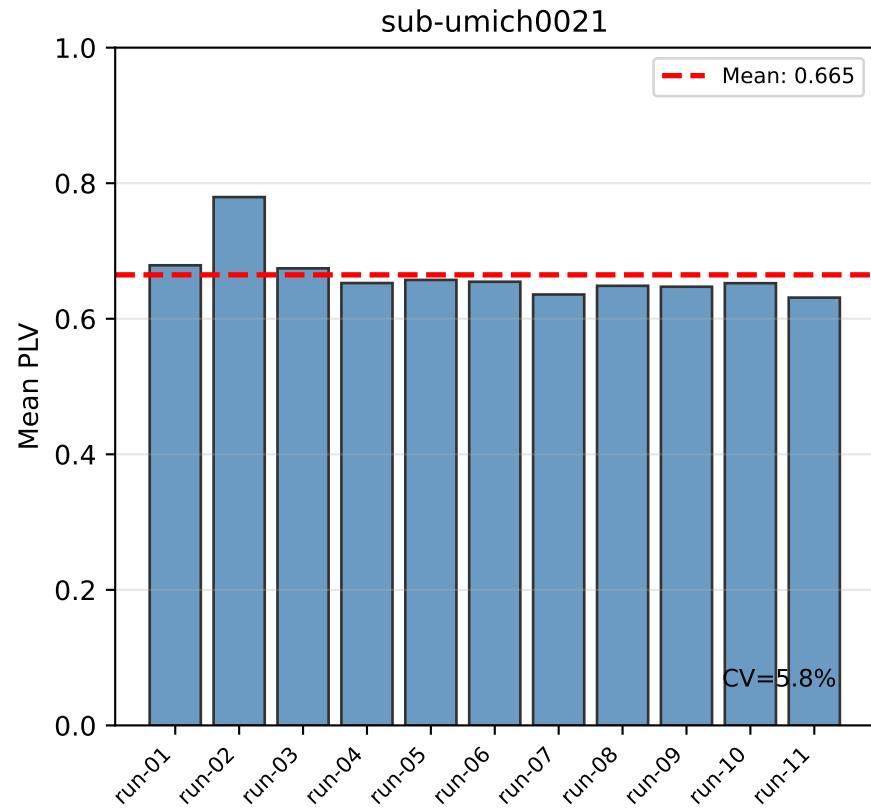
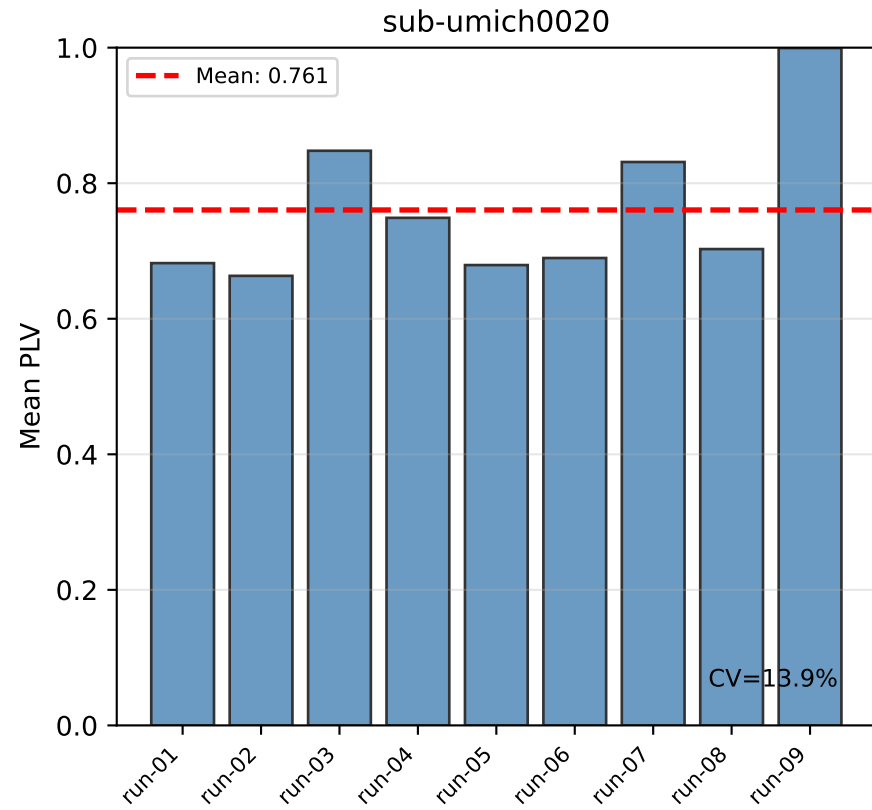


Figure 3: Distribution of Pairwise PLV Values

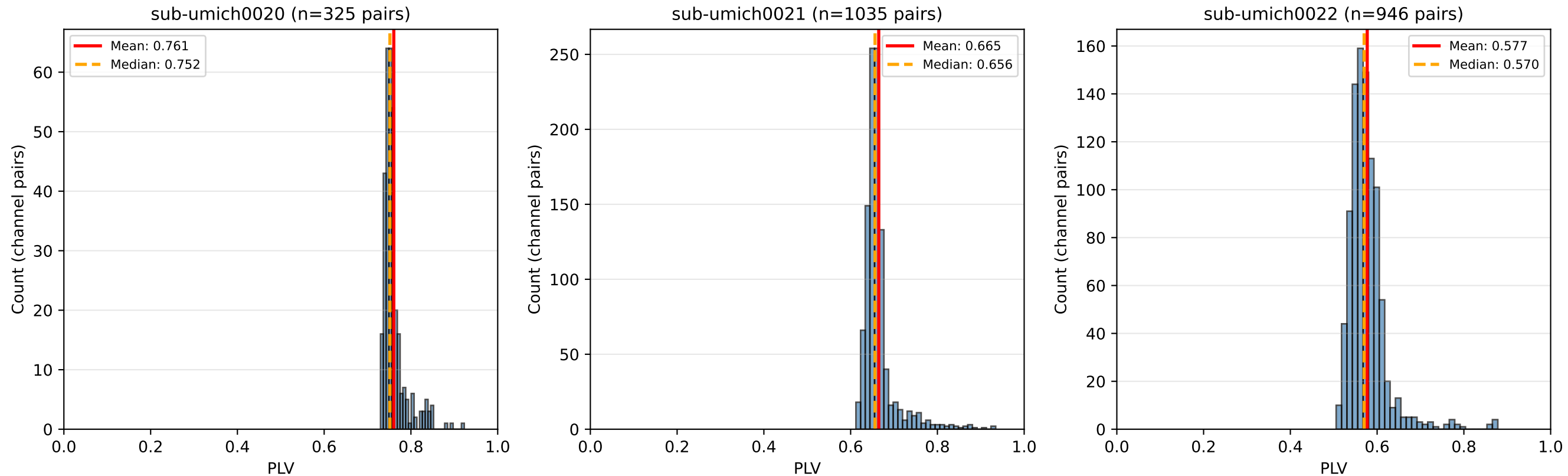
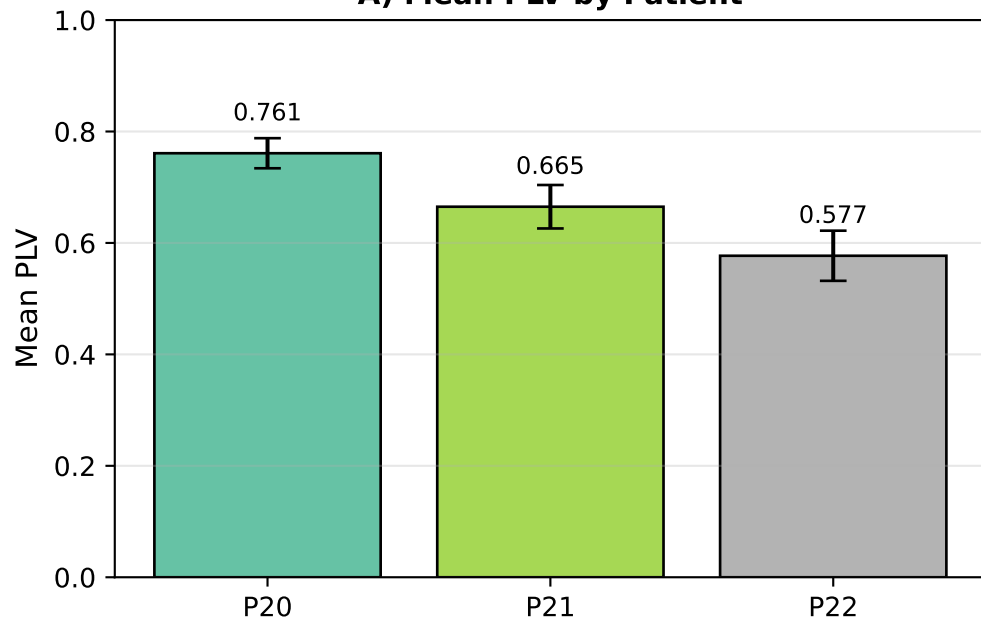
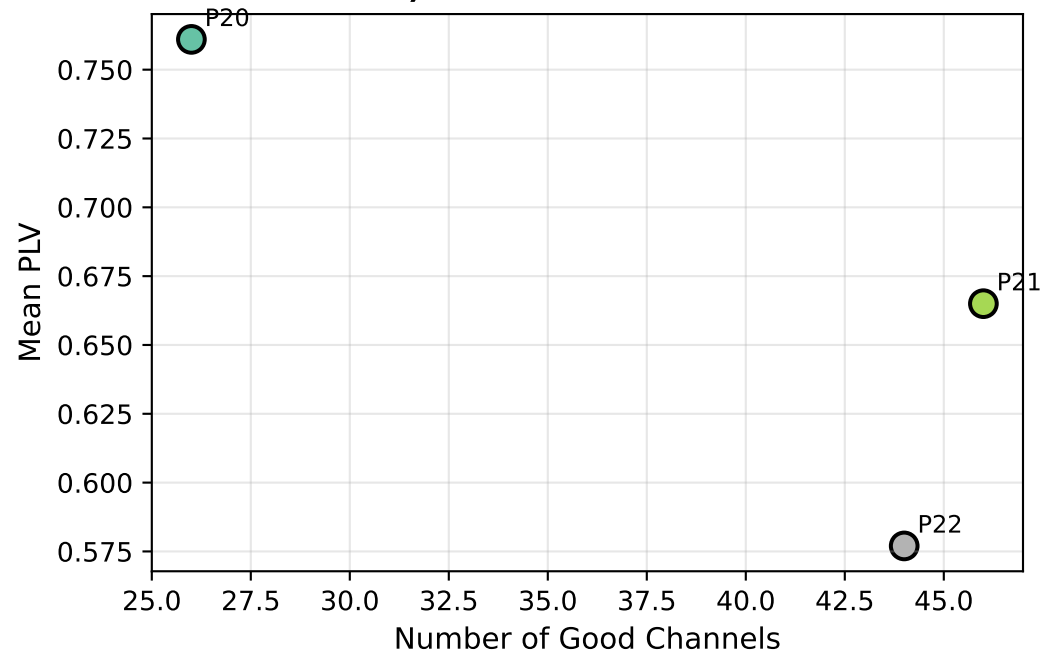


Figure 4: Cross-Patient Summary

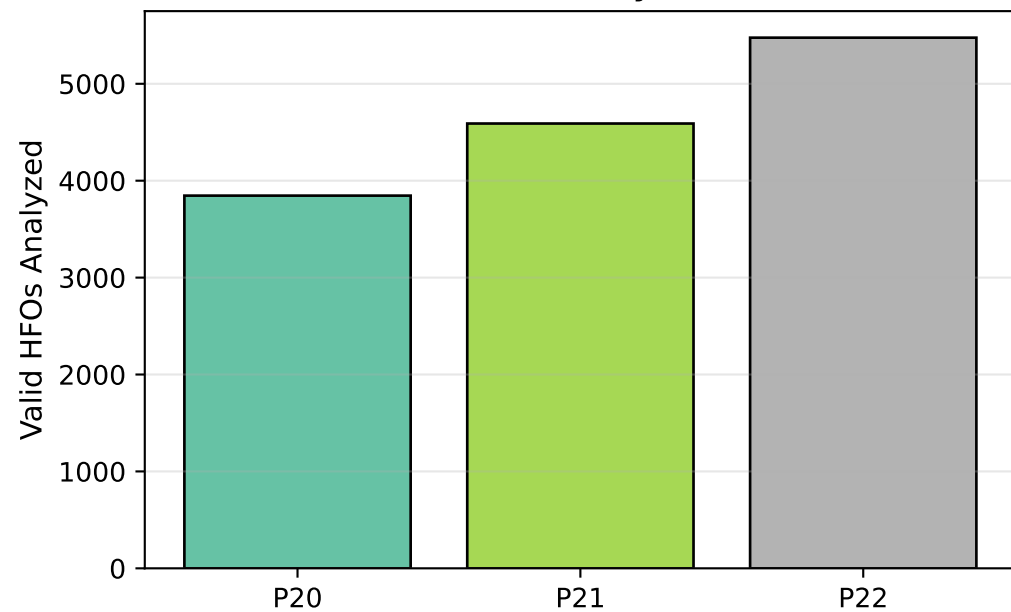
A) Mean PLV by Patient



B) PLV vs Channel Count



C) HFO Count by Patient



D) Recording Sessions by Patient

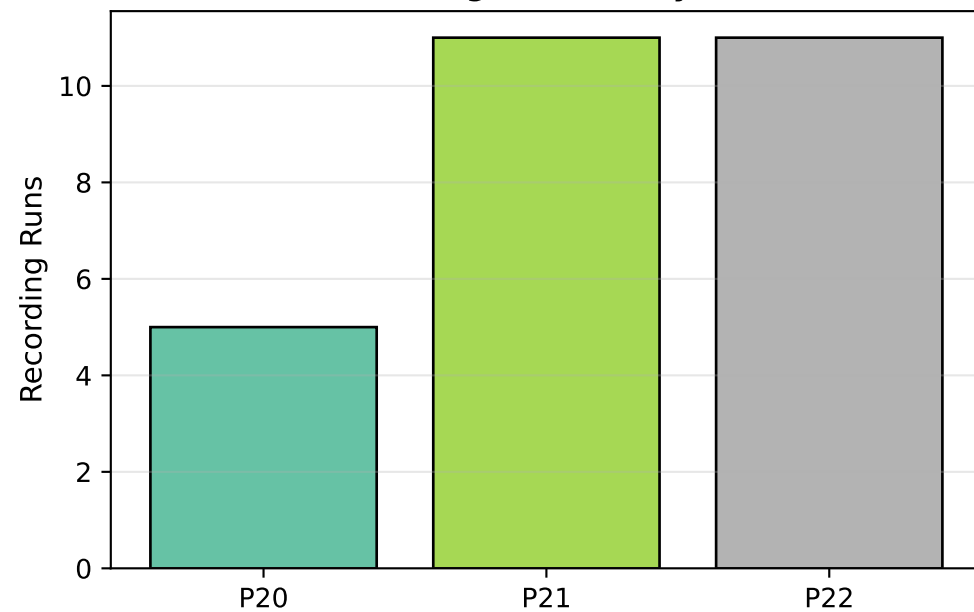
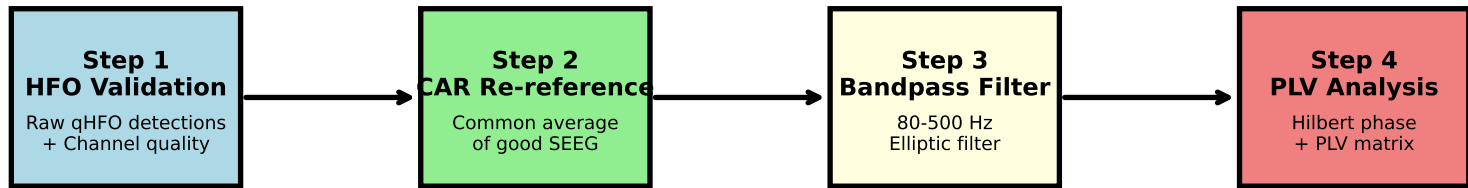


Figure 5: Analysis Pipeline Overview



Input:
H5 files
Channels.tsv

Output:
PLV matrices
Connectivity plots

Key Parameters:

- Sampling rate: 4096 Hz
- HFO band: 80-500 Hz
- Filter: Elliptic, order 10
- Max HFOs/run: 500
- Phase: Hilbert transform

$$PLV_{ij} = \left| \frac{1}{T} \sum_{t=1}^T e^{i(\phi_i(t) - \phi_j(t))} \right|$$

(Phase-Locking Value formula)

4. DISCUSSION

We have presented a Python-based pipeline for computing phase-locking values during high-frequency oscillation events in SEEG recordings. Our preliminary application to three epilepsy patients demonstrates that the approach is computationally tractable and yields quantifiable connectivity patterns.

4.1 Interpretation of PLV values

The mean PLV values we observed (0.53-0.76) are consistent with previous reports of phase synchronization in intracranial EEG. The substantial inter-individual variability is noteworthy and may reflect differences in pathology, electrode placement, or other factors.

4.2 Limitations

This study has several important limitations:

1. No statistical baseline - We did not compare HFO-period PLV to baseline periods or surrogate data. This is a critical gap.
2. Small sample size - Three patients is insufficient for clinical conclusions.
3. No clinical correlation - We did not examine relationships between PLV and seizure onset zone or surgical outcome.
4. Edge effects - The Hilbert transform produces unreliable phase estimates at segment edges, which we did not trim.
5. Volume conduction - High PLV between nearby contacts may reflect volume conduction rather than true neural coupling.

4.3 Future directions

Next steps include: (1) implementing surrogate-based statistical testing, (2) expanding to a larger cohort, (3) correlating PLV with clinical outcomes, and (4) comparing HFO-period PLV to non-HFO baseline periods.

5. CONCLUSION

We have developed and applied a Python pipeline for phase-locking value analysis during high-frequency oscillations in SEEG recordings. Preliminary findings from three patients suggest patient-specific PLV patterns with moderate within-subject stability. Substantial additional work is needed -- particularly statistical validation and clinical correlation -- before conclusions about clinical utility can be drawn.

REFERENCES

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Data and Code Availability: Pipeline available at GitHub (link to be added).
Raw data available upon request through data sharing agreements.